Project Title: - An Interactive Financial Data Warehousing and Data Mining Tool Using R

Github

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Problem Statement

Financial analysts frequently work with large Excel datasets that require repeated manual operations such as aggregation, visualization, and statistical analysis. These processes are time-consuming, error-prone, and lack automation.

There is a need for a system that:

- Accepts raw financial data
- Automatically structures it
- Allows flexible analysis
- Applies data mining techniques
- Generates insights dynamically

The main objective of this project is to design and develop an **interactive analytics tool using R and Shiny** that:

- Accepts large datasets (CSV / Excel)
- Performs **data preprocessing and transformation (ETL)**
- Enables **OLAP-based analysis**
- Applies **data mining techniques**
- Provides **interactive visualizations**
- Supports **AI-based query-driven analytics**

Dataset

- **Source**: Microsoft Financial Sample Dataset (commonly used for analytics and BI tasks)
- **Format**: Excel (.xlsx)
- **Number of Observations**: **711 records**
- **Number of Variables**: **16 attributes**

Attributes Description

Attribute	Description
Segment	Customer segment (Government, Midmarket, Enterprise, etc.)
Country	Country where sales occurred
Product	Product category
Discount Band	Discount level applied
Units Sold	Number of units sold
Manufacturing Price	Cost to manufacture product
Sale Price	Selling price per unit

Gross Sales	Total sales before discounts
Discounts	Discount amount applied
Sales	Net sales after discount
COGS	Cost of goods sold
Profit	Net profit
Date	Transaction date
Month Number	Month index (1–12)
Month Name	Name of month
Year	Year of transaction

Dataset Characteristics

- Contains both **categorical and numerical data**
- Includes **time-based attributes** → useful for time-series analysis
- Supports **multidimensional analysis**:
    - Region (Country)

    - Product
    - Segment
    - Time

Data Preprocessing Performed

- Removed missing values (`na.omit`)
- Standardized text fields (trim + title case)
- Converted `Date` column to proper date format
- Separated numeric and categorical variables for analysis

Why This Dataset is Suitable

This dataset is ideal for a **Data Warehousing and Data Mining project** because:

- Supports **OLAP operations** (multi-dimensional analysis)
- Contains **fact measures** (Sales, Profit, Units Sold)
- Includes **dimensions** (Time, Product, Country, Segment)
- Enables **data mining techniques**:

    Clustering (customer segmentation)

    Association rules (product relationships)

    Correlation analysis (financial metrics)

Methodology

1. Data Preprocessing (ETL)

- Data is loaded using:
    - `readxl` (Excel)
    - `data.table::fread` (CSV)
- Cleaning steps:
    - Missing value removal (`na.omit`)

    - String normalization (`str_trim`, `str_to_title`)
    - Date conversion
- Data is transformed into a structured format suitable for analysis

2. Data Warehousing Concepts

Although not stored in a physical warehouse, the system simulates:

- **Fact Data** → Numerical measures
- **Dimensions** → Categorical attributes

Supports:

- Grouping
- Aggregation
- Multidimensional analysis

3. OLAP Operations

Implemented dynamically:

- **Roll-up** → Aggregation (Sum, Mean, Count, etc.)
- **Drill-down** → Multi-level grouping
- **Slice** → Single dimension filtering
- **Dice** → Multi-dimensional filtering
- **Pivot** → Dynamic visualization

Implemented using:

- `dplyr::group_by`
- `summarise`
- dynamic UI controls

4. Data Mining Techniques

Correlation Analysis

- Uses `ggcorrplot`
- Identifies relationships between variables

Clustering (K-Means)

- Applied on selected numeric features
- Data is scaled before clustering
- Outputs:
  - Cluster groups
  - Visualization with ellipses

Association Rule Mining

- Uses `arules` package
- Finds frequent patterns and item relationships

AI-based Query Engine

- Uses external API (Google Gemini)
- Converts natural language queries into R code
- Executes dynamically on dataset

5. Visualization

Implemented using:

- `ggplot2`
- `plotly`

Features:

- Scatter plots
- Bar charts
- Line graphs
- Boxplots
- Interactive dashboards

6. Dashboard System (Shiny UI)

Built using:

- `shiny`

- `bs4Dash`

Modules:

- Dashboard summary
- Data explorer
- OLAP cube
- Visual analytics
- Correlation matrix
- Clustering
- Market basket analysis
- AI assistant

Results

The system successfully:

- Processes large datasets dynamically
- Performs real-time OLAP analysis
- Identifies correlations and patterns
- Segments data using clustering
- Extracts association rules
- Generates insights using AI-driven queries

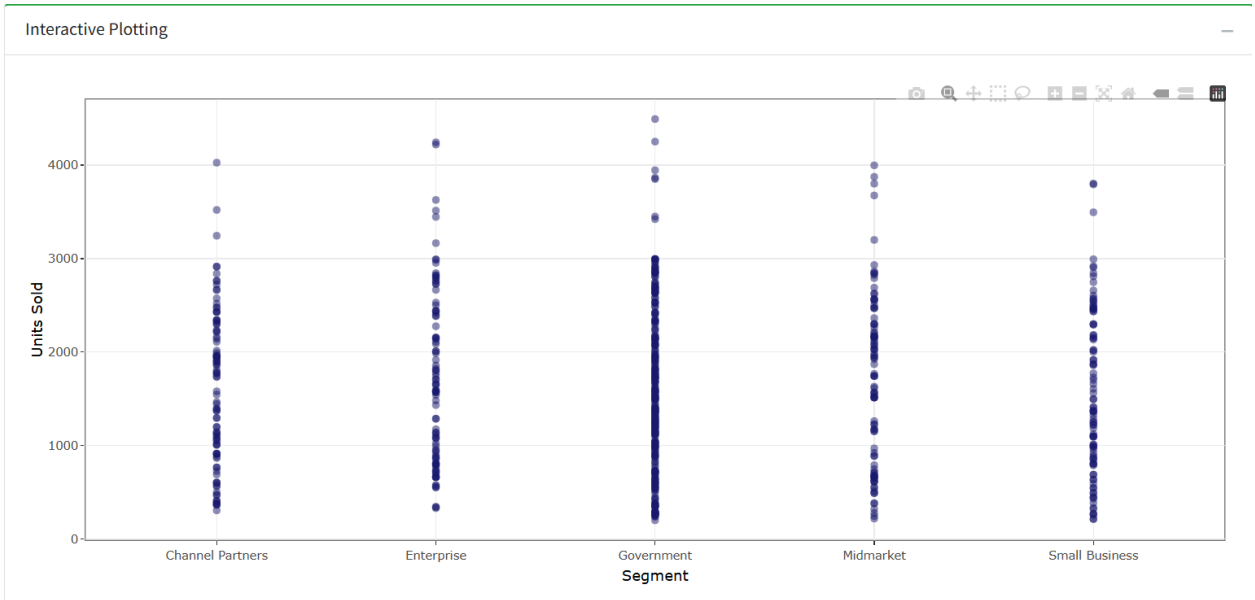
Key Observations:

- Numeric variables show strong correlation patterns
- Clustering reveals hidden group structures
- Association rules highlight frequent combinations
- AI assistant enables flexible analysis without coding

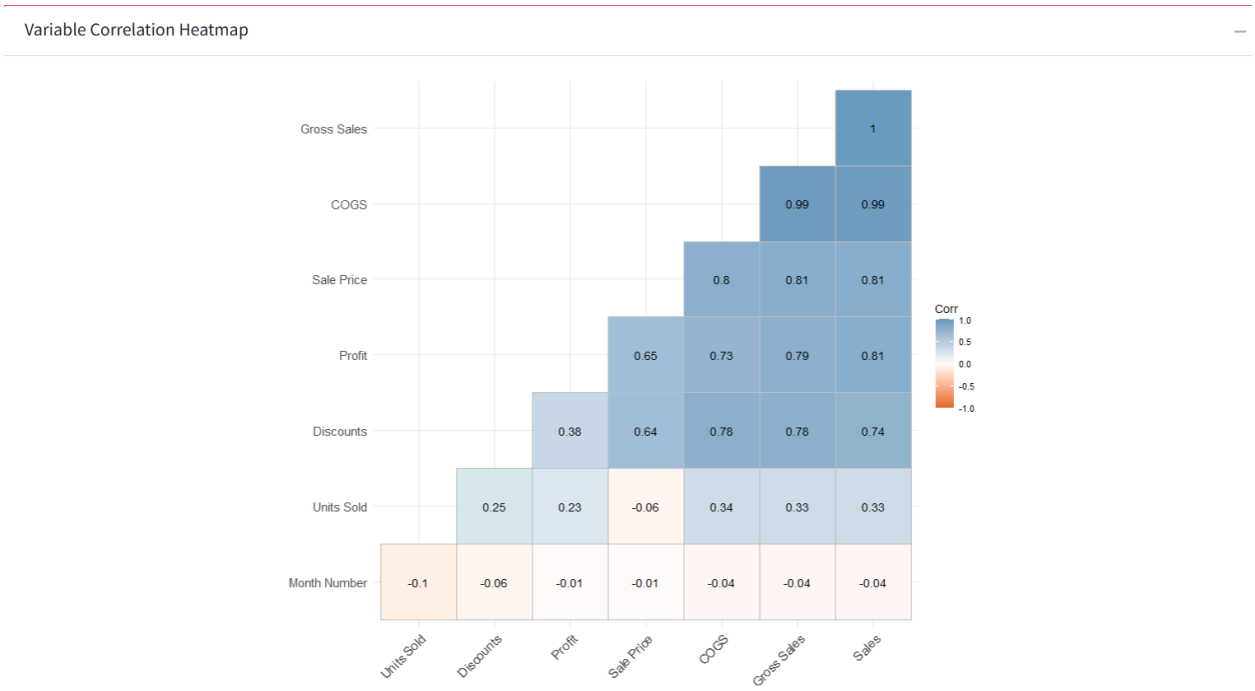
Key Visualizations

Generated dynamically:

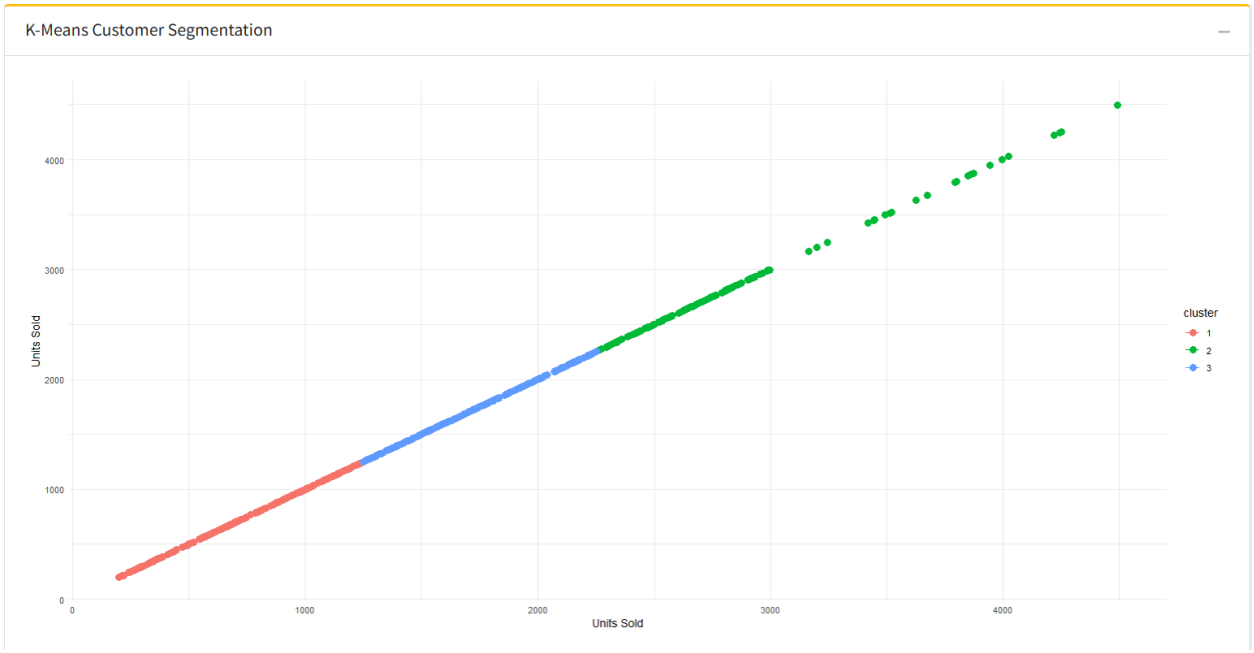
- Units Sold Variation Across Market Segments



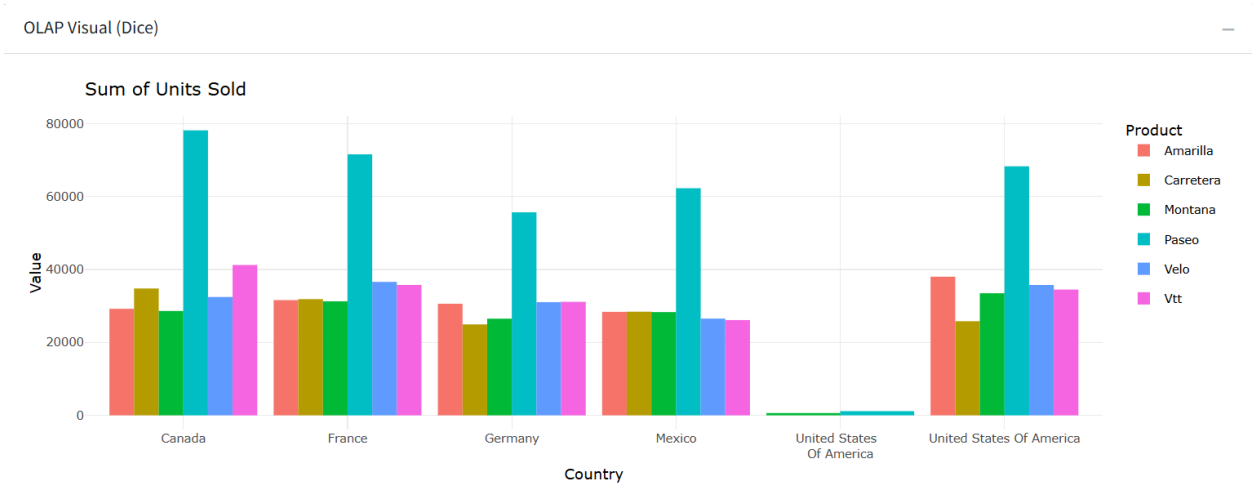
- Correlation heatmap



- Cluster visualization



- OLAP aggregation chart



How to Run the Project

1. Install Required Packages

install.packages(c(

"shiny","bs4Dash","readxl","data.table","dplyr",

```
 "ggplot2","DT","httr","jsonlite","ggcorrplot",
 "plotly","arules","stringr"
))
```

2. Run the Application

```
shiny::runApp("app.R")
```

3. Usage Steps

1. Upload dataset
2. Navigate tabs:
    - Data Explorer
    - OLAP Cube
    - Visualization
    - Mining modules
3. Use AI Assistant for custom queries

Folder Structure

```
project-repository/
│── README.md
│── data/
│   └── dataset_description.md
│── scripts/
│   ├── 01_data_preparation.R
│   ├── 02_exploratory_analysis.R
│   ├── 03_modeling.R
│   ├── 04_evaluation.R
│── app/
│   └── app.R
│── results/
│   ├── figures/
│   └── tables/
│── presentation/
│   └── project_presentation.pptx
```

R Libraries Used

- `shiny`, `bs4Dash` → UI framework
- `dplyr`, `data.table` → data manipulation

- `ggplot2`, `plotly` → visualization
- `ggcorrplot` → correlation
- `arules` → association mining
- `httr`, `jsonlite` → API integration
- `stringr` → text processing

Conclusion

This project successfully demonstrates the integration of:

- **Data Warehousing concepts** (ETL, OLAP)
- **Data Mining techniques** (clustering, association rules, correlation)
- **Interactive analytics** using R

The system provides a **powerful, flexible, and automated financial analysis tool** that reduces manual effort and enhances decision-making capabilities.

Contribution

- **Vadan Datta** → Data mining modules
- **Hrishith Reddy** → UI & dashboard design, Core architecture
- **Balaji Bhargav** → AI module, OLAP logic
- **Srujana** → Testing & documentation

References

Dataset :- https://www.kaggle.com/datasets/luckyrao/financial-sample-data